Problem 1.

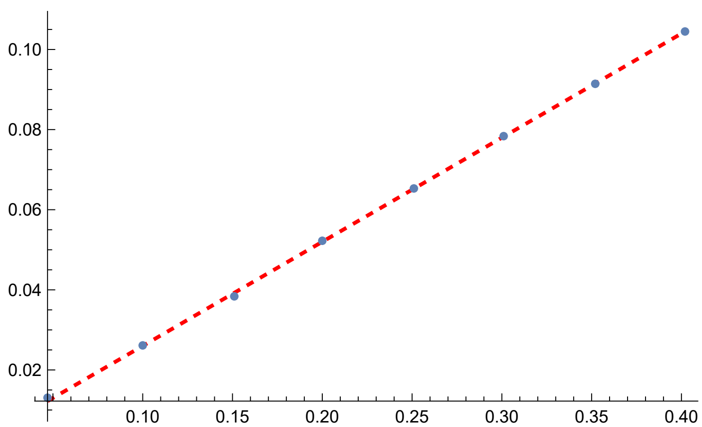
In[◦]:=
```
KE = {16, 32, 47, 64, 80, 96, 112, 128} / 1000;
d = {0.047, 0.10, 0.151, 0.200, 0.251, 0.301, 0.352, 0.402};
m = 0.125; g = 9.80;
```

In[◦]:=
```
y = KE/(m*g);
data = Thread[{d, y}];
dataplot = ListPlot[data];
fit = LinearModelFit[data, X, X, IncludeConstantBasis → False];
fit["ParameterTable"]
fitplot = Plot[fit[X], {X, Min[d], Max[d]}, PlotStyle → {Red, Dashed}];
Show[fitplot, dataplot]
```

Out[◦]=

	Estimate	Standard Error	t-Statistic	P-Value
X	0.259933	0.000666788	389.828	1.93019×10^{-16}

Out[◦]=



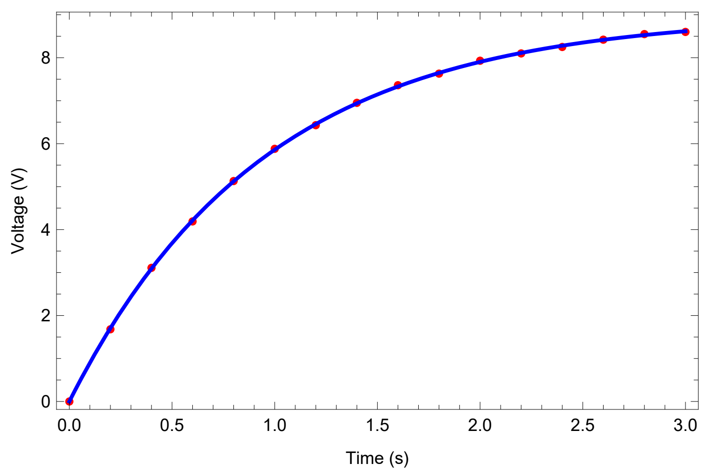
Problem 4.
Part (a)

In[51]:=
```
tVals = {0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0, 2.2, 2.4, 2.6, 2.8, 3.0};
VcVals = {0.00, 1.68, 3.11, 4.19, 5.13, 5.88,
   6.43, 6.95, 7.36, 7.63, 7.93, 8.10, 8.25, 8.42, 8.55, 8.60};
resistor = 22600;
data = Transpose[{tVals, VcVals}];
nlm =
  NonlinearModelFit[data, v * (1 - Exp[-t / (resistor * cap)]), {{v, 9.0}, {cap, 0.0001}}, t];
(*results*)
nlm["ParameterTable"]
Show[ListPlot[data, PlotStyle → Red, Frame → True,
  FrameLabel → {"Time (s)", "Voltage (V)"}], Plot[nlm[t], {t, 0, 3}, PlotStyle → Blue]]
```

Out[56]=

	Estimate	Standard Error	t-Statistic	P-Value
v	8.99831	0.016171	556.447	8.09032×10^{-32}
cap	0.0000420104	2.09302×10^{-7}	200.716	1.27901×10^{-25}

Out[57]=



Problem 4(b)

In[122]:=

```
current = Table[(VcVals⟦i + 2⟧ - VcVals⟦i⟧) / (2 * 0.2), {i, 1, Length[VcVals] - 2}];
t2 = Table[tVals⟦i⟧, {i, 1, Length[VcVals] - 2}];
idata = Thread[{t2, current}];
ifit = NonlinearModelFit[idata, a * Exp[-k * t], {{a, 10}, {k, 0.5}}, t];
Print[ifit["ParameterTable"]]
Show[ListPlot[idata, PlotStyle → Green,
  Frame → True, FrameLabel → {"Time (s)", "dV/dt (V/s)"}],
 Plot[ifit[t], {t, 0.4, 2.6}, PlotStyle → Black]]
```

	Estimate	Standard Error	t-Statistic	P-Value
a	7.76602	0.0650122	119.455	7.94181×10^{-20}
k	1.05614	0.0143872	73.4084	2.71684×10^{-17}

Out[127]=